Industrial Internship Report on

” Password Generator System ”

Prepared by -

Ankur Ghosh

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My password generator project was a good learning experience. It helped me understand the use of Python concepts like the random module and string handling. I learned how to create secure passwords and improved my coding and problem-solving skills. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

- Summary of the whole 6 weeks' work –

During the 6 weeks, I learned the basics of Python programming, including variables, data types, conditional statements, loops, and functions. I also gained knowledge of NumPy and its operators. Along with theory, I practiced small programs and completed a password generator project, which helped me apply my learning practically.

- About need of relevant Internship in career development –

Such training is important for career development as it helps in building practical skills, improving problem-solving ability, and understanding real-world applications. It also increases confidence and prepares students for future job opportunities.

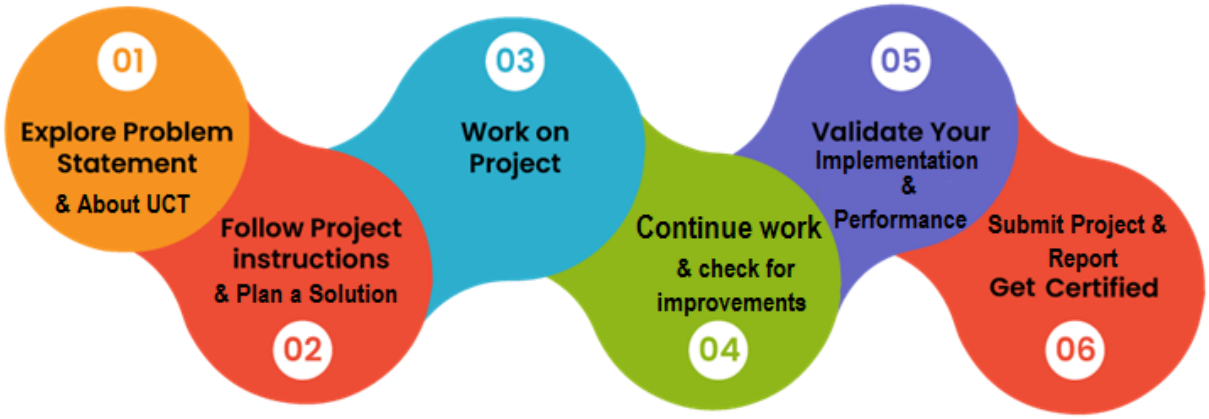
- Brief about Your project/problem statement –

The project was a password generator, which creates strong and secure passwords using a combination of letters, numbers, and special characters. The main objective was to understand how to generate random and secure data using Python.

- Opportunity given by USC/UCT –

The program provided a good opportunity to learn Python in a structured way. It offered learning resources, practical tasks, and guidance that helped in improving programming skills and understanding real-world applications.

- How Program was planned -



- Your Learnings and overall experience –

Overall, the experience was very helpful and informative. I improved my programming knowledge, logical thinking, and confidence in coding. The learning process was smooth, and the practical approach made it more interesting and effective.

Thanks to all who have helped you directly or indirectly.

- Your message to your juniors and peers –

I would like to encourage my juniors and peers to actively participate in such learning programs. These experiences help in building strong basics and improving practical skills. Stay consistent, practice regularly, and don't hesitate to ask questions. With dedication and curiosity, you can gain valuable knowledge and grow in your career.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

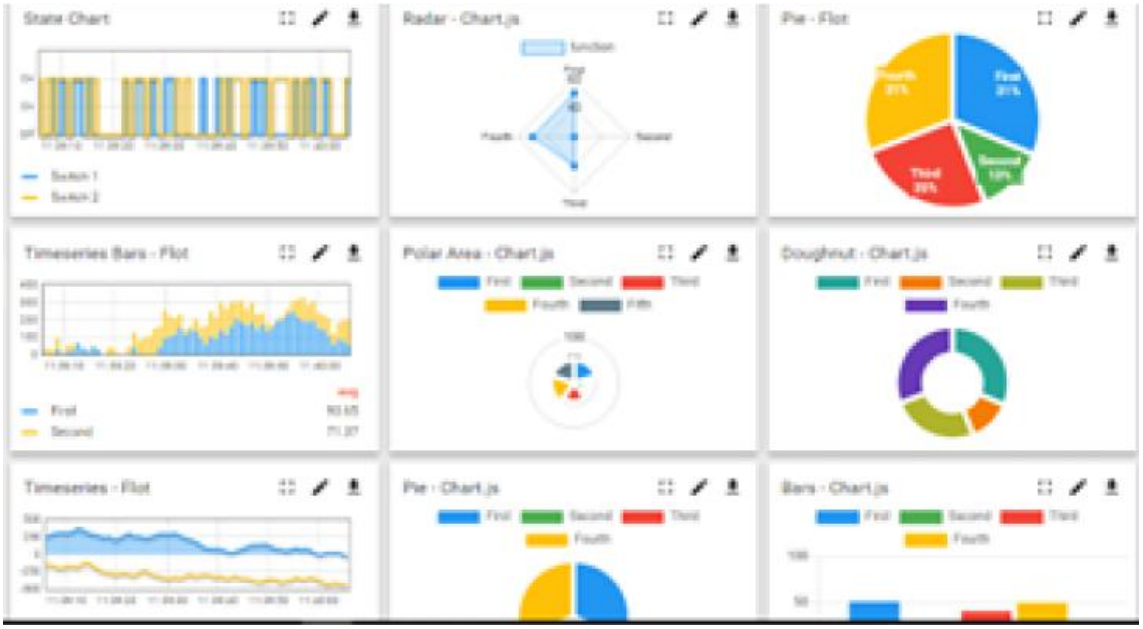
i. UCT IoT Platform (UCT Insight)

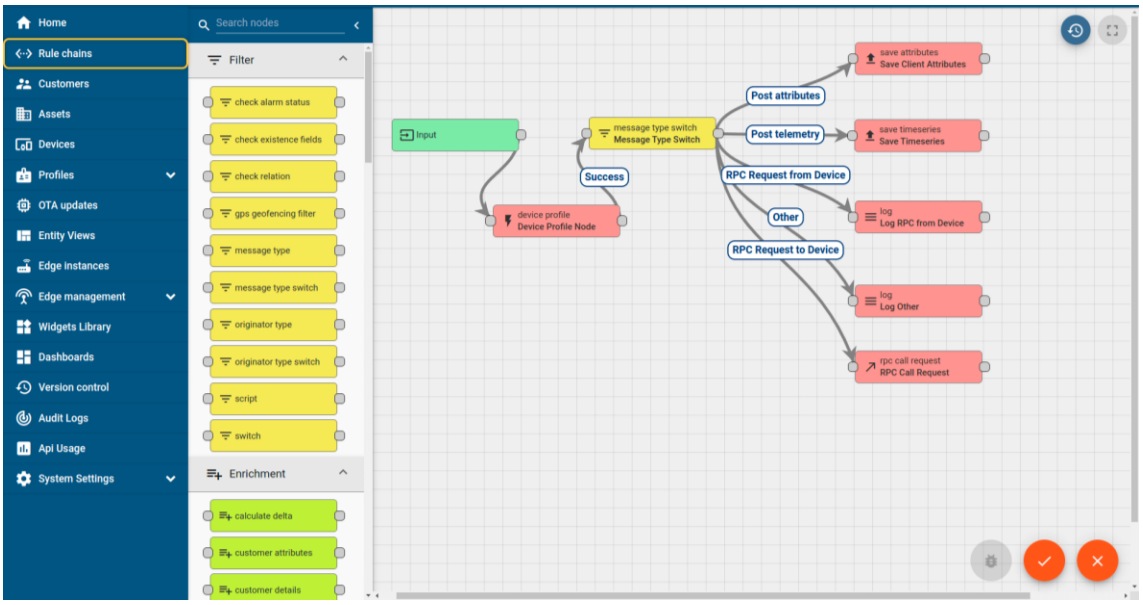
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various MySQL Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application (Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and help to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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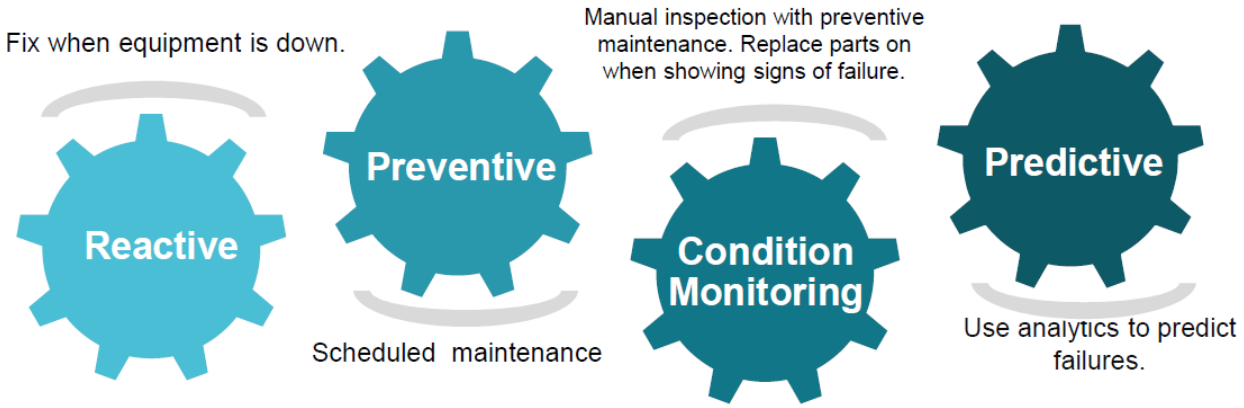
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

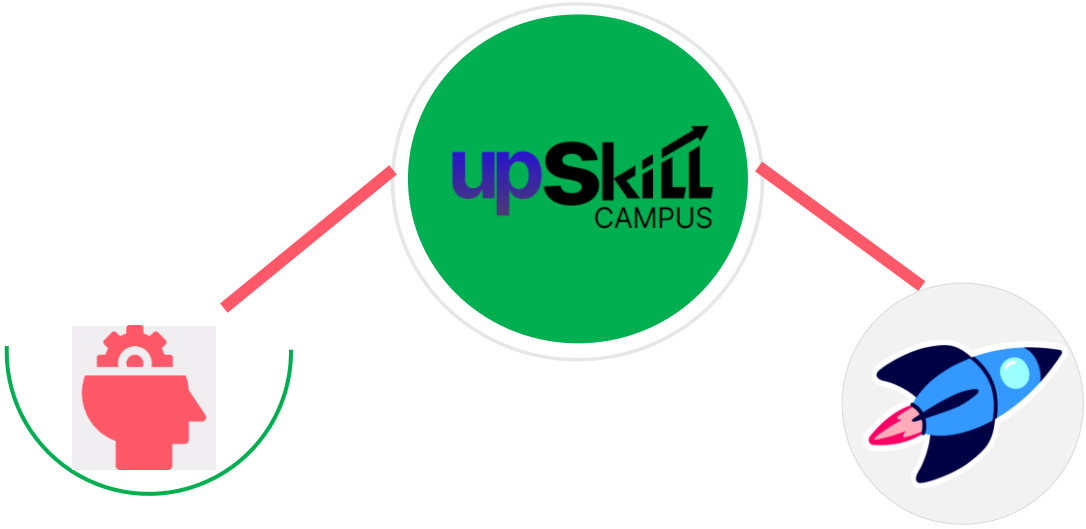
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

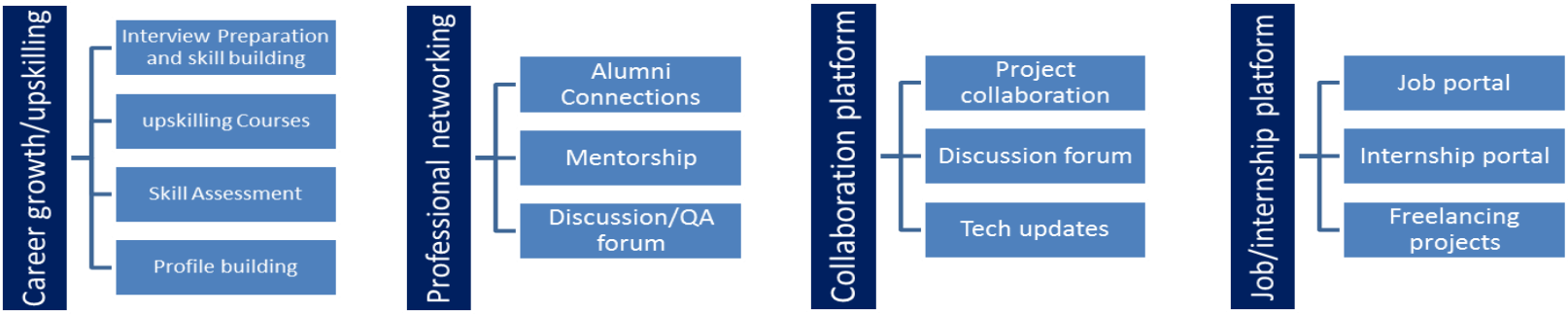
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that runs long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

3 Problem Statement

The objective was to develop a **Password Generator using Python**. The problem focused on creating strong and secure passwords by combining letters, numbers, and special characters. The goal was to ensure that the generated passwords are random and difficult to guess, improving security for users. This project helped in understanding how to apply Python concepts like randomization and string handling in a real-world scenario.

- Initially confused about differences between Python lists and NumPy arrays.
- Needed additional practice to correctly apply NumPy operators on arrays.
- Understanding element-wise operations required careful observation.

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

What is your proposed solution?

What value addition are you planning?

4.1 Code submission (Github link)

https://github.com/ankur2710/Upskillcampus-/blob/main/Gui_password_generator.py

4.2 Report submission (Github link)

5 Proposed Design/ Model

The solution is designed to generate secure passwords using Python by following a step-by-step process. The program starts by taking user input such as the desired password length and type of characters (letters, numbers, special characters). In the next stage, it uses Python libraries like random and string to select characters randomly. These characters are then combined to form a password. Finally, the generated password is displayed to the user.

The design ensures that the password is random, secure, and meets user requirements.

5.1 High Level Diagram

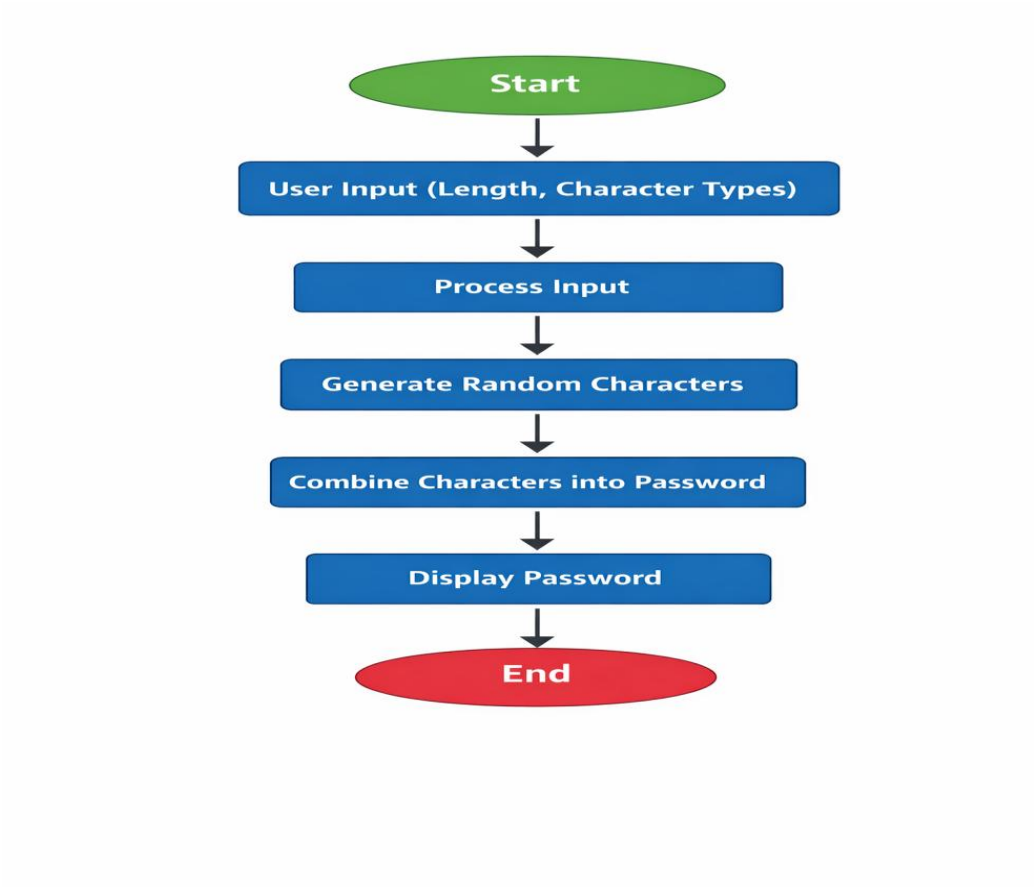
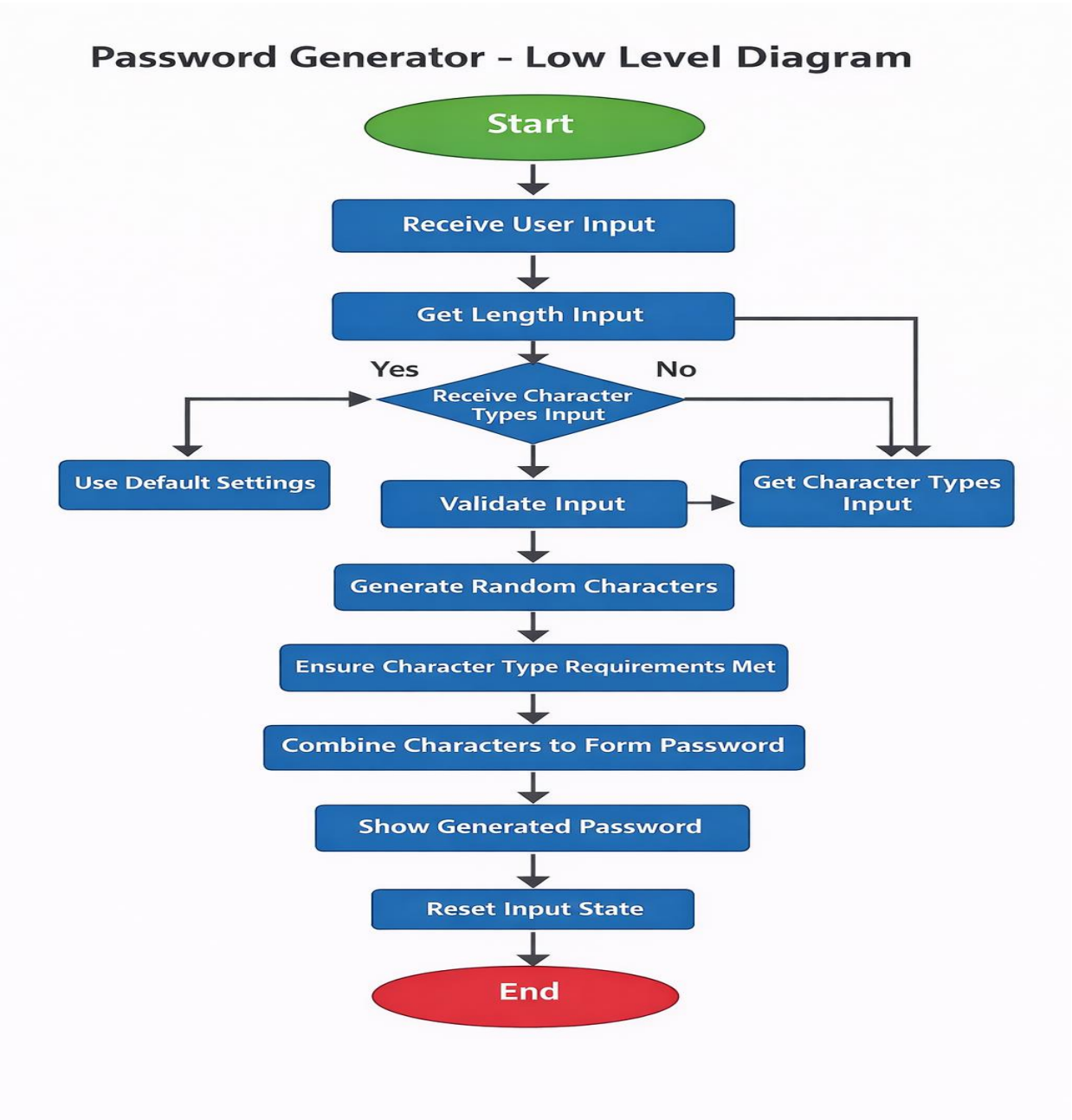
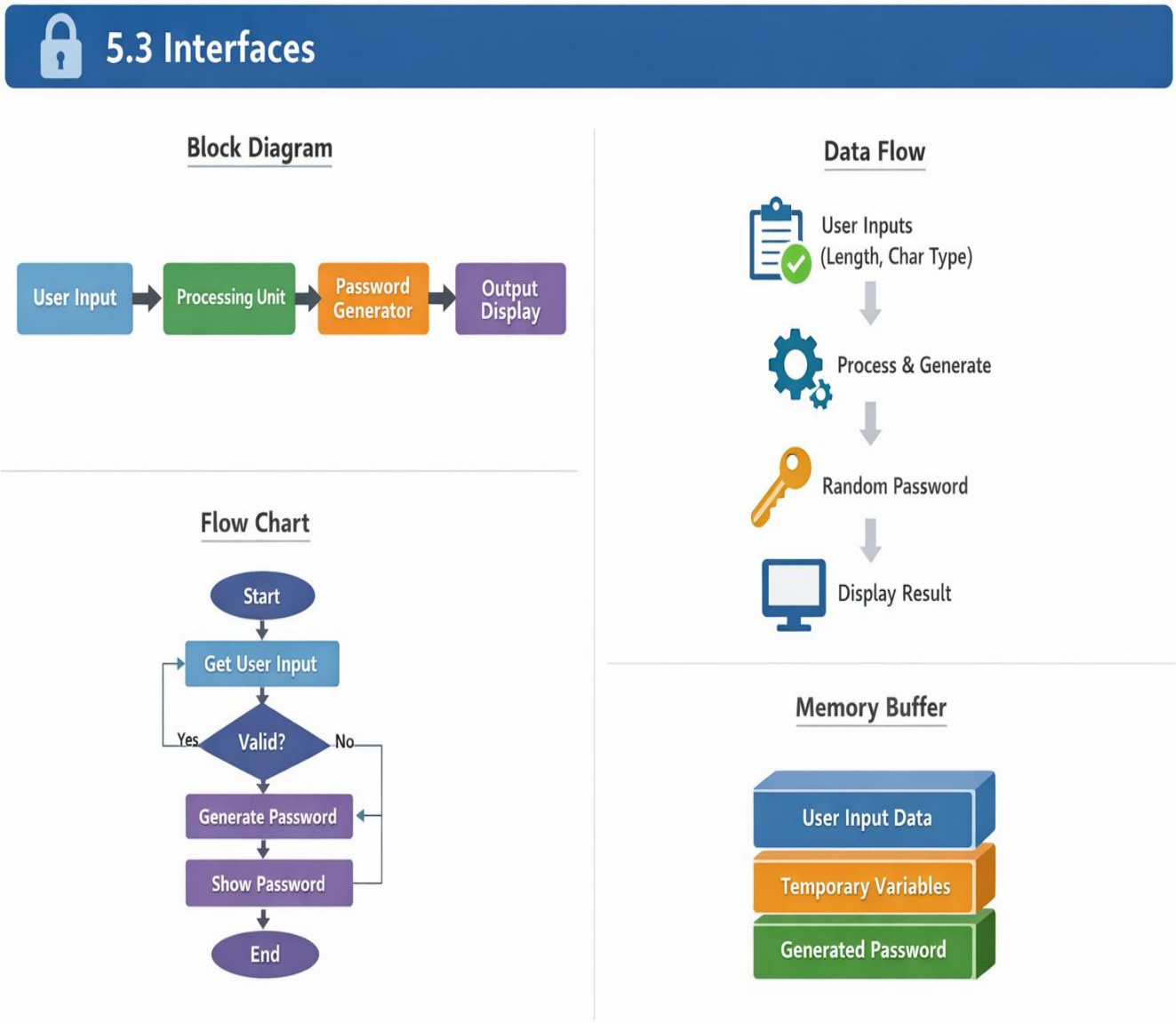


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram



5.3 Interfaces



6 Performance Test

The performance of the Password Generator was evaluated based on constraints such as speed, memory usage, and randomness of generated passwords. The program is lightweight and executes quickly, making it suitable for real-world usage. Memory consumption is minimal as only temporary variables are used during execution.

The main constraint was ensuring randomness and security of passwords. This was handled using Python's random module, which generates unpredictable combinations of characters. The design ensures that the generated passwords are strong and difficult to guess.

6.1 Test Plan/ Test Cases

- Test with different password lengths (e.g., 6, 8, 12 characters)
- Test with different combinations (letters only, letters + numbers, all characters)
- Test for repeated execution to check randomness
- Test invalid inputs (negative length or empty input)

6.2 Test Procedure

- Run the program and enter different input values
- Observe the generated password for correctness
- Repeat execution multiple times to check variation
- Verify that the output meets the required conditions

6.3 Performance Outcome

The program successfully generated secure and random passwords for all test cases. It performed efficiently with fast execution and low memory usage. The results confirmed that the system works correctly and meets the required functionality for generating strong passwords.

7 My learnings

During this project and the overall training period, I gained a strong understanding of Python programming concepts such as variables, data types, conditional statements, loops, and functions. I also learned how to use basic libraries like NumPy and random, which helped me understand how Python can be applied to solve real-world problems. Working on the password generator project gave me practical experience in implementing these concepts.

This journey also improved my logical thinking and problem-solving skills. Initially, I found it challenging to apply logic in programs, but with continuous practice, I became more confident in writing code and debugging errors. I learned how to break down a problem into smaller steps and solve it systematically.

I understood the importance of writing clean and readable code. I also became familiar with testing my programs using different inputs to ensure correctness. This helped me develop a more structured approach towards programming.

This experience has been very valuable for my career growth. It has built a strong foundation in programming and increased my confidence to work on more advanced topics in the future. The skills I gained during this period will help me in academic projects as well as in pursuing opportunities in the IT and software development field.

8 Future work scope

The current password generator system can be further enhanced by incorporating advanced features to improve usability, security, and scalability. One significant improvement would be the development of a graphical user interface (GUI), enabling a more user-friendly and interactive experience compared to the existing command-line approach.

Additionally, the system can be extended to include password strength evaluation mechanisms, which analyse and rate the generated passwords based on security standards. Implementing encryption techniques for secure storage of passwords is another important enhancement that would make the application more reliable for real-world usage.

Further improvements may include customizable password policies, such as enforcing specific rules for length, uppercase letters, numbers, and special characters. The project can also be expanded by integrating it with web or mobile platforms, making it more accessible and practical. These enhancements would significantly increase the applicability of the system in real-world scenarios and align it with industry standards.